33kV OVERHEAD LINE FROM PARKSTON
POWER STATION POLE FIM FA35 RECLOSER
2000-F12-SLD-EL-10001

F12-TR-012
33 / 11kV
5000kVA
Dyn11
z=7%
KNAN

F12-NR-012
PRIMARY CRUSHING 33/11kV
TRANSFORMER F12-TR-012
NEUTRAL EARTHING RESISTOR
2000-EL011-B02-10004

30/0.05A
EFCT

25A FOR 10s
254Ω +/-5%

SENSING
ER-15kV

240V SUPPLY
F12-DB-001 SWITCHROOM L&SP DB
2000-F12-TLD-EL-10101

F12DB001-R12
F12DB001-N12
HEATER

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD PANEL A02
2000-EL003-F16-10005

EARTH BAR
QCE
BAD
PFV1
RD

BCT1 300 / 1A 2.5VA CL 0.2
BCT2 300 / 1A 2.5VA CL 0.2
BCT3 300 / 1A 2.5VA CL 0.2
SEL 735

BCT1 300 / 1A 2.5VA 5P20
BCT2 300 / 1A 2.5VA 5P20
BCT3 300 / 1A 2.5VA 5P20
SEL 751

2000-EL003-F16-10005
SHEET 16

-QAB
630A

RD
WH
BU
11kV 1250A 3-E 25kA FOR 3s

BCF2
(PART OF)
CONSTRUCTION NOTE 2

F12TR012-JB
PRIMARY CRUSHING TRANSFORMER F12-TR-012 PROTECTION DEVICES TERMINAL BOX
2000-EL011-D02-10001

PRV
PRESSURE RELIEF VENT TRIP ALARM
OIL LEVEL
OIL LEVEL ALARM (MINIMUM)
OIL LEVEL
OIL LEVEL ALARM (MAXIMUM)
OIL TEMP
OIL TEMPERATURE ALARM (100°C)
OIL TEMP
OIL TEMPERATURE TRIP (110°C)
WINDING
WINDING TEMPERATURE ALARM (110°C)
WINDING
WINDING TEMPERATURE TRIP (125°C)

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD PANEL A02
2000-EL003-F16-10005

BCF1 (PART OF)
1535 X2 10 — 48VDC CONTROL — D10 IN409 TRANSFORMER PROTECTION TRIP
1545 X2 55
1595 X2 14 — 48VDC CONTROL — D14 IN413 TRANSFORMER OIL LEVEL MAXIMUM ALARM
1605 X2 59
1625 X2 15 — 48VDC CONTROL — D15 IN414 TRANSFORMER OIL TEMPERATURE ALARM
1635 X2 60
1685 X2 16 — 48VDC CONTROL — A10 IN101 TRANSFORMER WINDING TEMPERATURE ALARM
1695 X2 61
SHEET 13

F12SR001-LCS
SWITCHROOM EMERGENCY SHUTDOWN
PUSHBUTTON RELAY K2 (LOCATED IN AIR LOCK)
2000-F12-TER-EL-10202

1760 X3 23
1780 X3 33
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON CONNECTION TO THE VCB 1st TRIP CIRCUIT SHEET 8

E-SHUT
NOTE 5

1800 X5 32
X5 34
F12SR001LCS-48VDC+
F12SR001LCS-48VDC-
X5 50
X5 51
UPSTREAM SHUTDOWN MARSHALLIN TERMINALS

BCF1 (PART OF)
C03 OUT302
C04
PROTECTION TRIP UPSTREAM SHEET 15

F12-BC-001
11kV SWITCHROOM
48VDC BATTERY CHARGER
2000-F12-TLD-EL-10001

POLE FA35 RECLOSER
(NOTE 4)
K1
RECLOSER INTERPOSING TRIP RELAY
1915 IN ON
1935 IN 1+ INPUT 1 — DOWNSTREAM SWITCHROOM INTERTRIP
CONSTRUCTION NOTE 1
IN 2+ INPUT 2
IN 3+ INPUT 3
I/O MODULE
2000-CN008-D02-30001

F12-DCS-401
415V SWITCHROOM DCS PANEL
2000-F12-TER-IC-10014

X2 30 F12SW012OP+48VDC — R1 11 — 1A — 01/03/01 — 1B — DCS REMOTE CB OPEN
X2 63 F12SW012OP — R1 14
X2 31 F12SW012CL+48VDC — R2 11 — 2A — 01/03/02 — 2B — DCS REMOTE CB CLOSE
X2 64 F12SW012CL — R2 14

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F12SW012-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. ALL ALARMS AND TRIPS SHALL BE FAIL-SAFE. NOTE THAT THE PROTECTION RELAY INPUTS SHALL BE INVERTED, I.E. ALARM AND TRIP WHEN CONTACT IS OPENED.
4. NOTE THAT CROSS SECTIONAL AREA OF CABLES TO THE OVERHEAD LINE RECLOSER ARE INCREASED IN SIZE - REFER TO THE CABLE SCHEDULE.
5. E-SHUT CIRCUIT IS FAIL SAFE. NORMALLY CLOSED CONTACTS WILL BE OPEN WHEN THE E-SHUT BUTTON IS UN-ACTIVATED AND THE CIRCUIT HEALTHY.

CONSTRUCTION NOTES

1. INSTALL NEW RELAY K1 AND UTILISE SPARE X5 TERMINALS FOR TRIPPING UPSTREAM 33kV OHL RECLOSER.

   INSTALL NEW RELAY K1 INTO POLE FA35 RECLOSER CONTROL PANEL.

   CONFIRM TERMINALS X5:50 & X5:51 ARE SPARE.

   TERMINATE NEW 2C+E CABLE IN K1:A1 AND K1:A2 TERMINALS.

   INSTALL WIRING FROM K1 N/O CONTACT TERMINAL 33 TO INPUT 1 OF I/O MODULE AND WIRING FROM K1 N/O CONTACT TERMINAL 34 TO COM OF I/O MODULE.

   REFER TO ABB DRAWING 2000-EL003-D03-10001 SHEET 4 FOR X5 TERMINALS.
2. CONFIRM TERMINAL X4-23 IS SPARE.
   INSTALL WIRING FROM TERMINAL X4-23 TO (SE330 AU) BCF2 TERMINAL R:6.

LEGEND

- xx/xx/xx — DCS DIGITAL INPUT
- xx/xx/xx — DCS DIGITAL OUTPUT
- xx/xx/xx — DCS ANALOGUE INPUT
- xx/xx/xx — DCS ANALOGUE OUTPUT
- xx/xx/xx — DCS ANALOGUE RTD INPUT
- ⊗ — 11kV SWITCHBOARD TERMINAL 1
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- (x) — DEVICE TERMINAL

PANEL A02

2000-F12-SCM-EL-10103_Rev1

F12-SW-012 incomer schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235I7 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F12 - PRIMARY CRUSHING\2000-F12-SCM-EL-10103.DWG 13AUG25 11:13AM HOTSEAT04